

presolve
↓
root node LP
↓
cutting planes
↓
branch & bound

Practitioner

(trial & error)

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best practices for modelling



Practitioner

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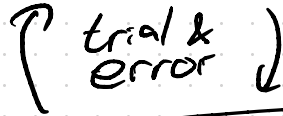


branch & bound

best practices for modelling



Practitioner



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branch & bound

Solver capabilities
↕
Solver usage

Sources of high condition numbers

- Remedy: calculate your data in double or higher precision to limit round-off errors
- Use integral data wherever possible

bad: $0.3333333 \ x + 0.6666666 \ y = 1$

better: $0.3333333333333333 \ x + 0.6666666666666666 \ y = 1$

best: $1x + 2y = 3$

- Above coefficients are clearly meant to be 1/3 and 2/3, but what about numbers like 0.3823529412 or 0.17525773195876287?
- See https://en.wikipedia.org/wiki/Repeating_decimal for common truncated decimal representations
- `Fraction.limit_denominator()` function from Python fractions package provides fractional representation of repeating decimal representations.



Pinpointing Numerical Issues: Tool for Diagnosis & Explanation of Ill Conditioning in LPs & MILPs